



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Discover. Learn. Empower.

Experiment -8

Student Name: Anadi Mishra

UID:22BCS11426

Branch: BE-CSE

Section/Group:IOT_638-B

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Subject Name: Project-Based Learning **Subject Code:** 22CSH-359 in Java with Lab

7.1.1.Aim: To develop a servlet that accepts user credentials from an HTML form and displays a personalized welcome message on successful login.

7.1.2 Objective: Learn form handling with Servlets
Understand HTTP request/response handling Practice
doPost() method

7.1.3 Code:

```
<!DOCTYPE html>
<html>
<head><title>Login</title></head>
<body>
  <form action="LoginServlet" method="post">
    Username: <input type="text" name="username"><br>
    Password: <input type="password" name="password"><br>
    <input type="submit" value="Login">
  </form>
</body> </html>
```

```
import java.io.*; import
javax.servlet.*; import
javax.servlet.http.*;
```

```
public class LoginServlet extends HttpServlet { protected void
doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
    String user = request.getParameter("username");
    String pass = request.getParameter("password");
```

```
response.setContentType("text/html"); PrintWriter
out = response.getWriter(); if ("admin".equals(user)
&& "1234".equals(pass)) {
out.println("<h2>Welcome, " + user + "!</h2>");
} else { out.println("<h2>Login Failed. Invalid
credentials.</h2>"); }
}
```

```
<web-app>
<servlet>
<servlet-name>LoginServlet</servlet-name>
<servlet-class>LoginServlet</servlet-class>
</servlet>
<servlet-mapping>
<servlet-name>LoginServlet</servlet-name>
<url-pattern>/LoginServlet</url-pattern>
</servlet-mapping>
</web-app>
```

Output:

- 1) On correct login: Welcome, Rithika!
- 2) On failure: Login Failed. Invalid credentials.

7.2.1 Aim: To build a servlet integrated with JDBC that displays all employees and enables search by employee ID.

Objective: 1) Use JDBC with Servlet

2) Fetch and display records

3) Implement search functionality

7.2.2 Code:

```
<!DOCTYPE html>
<html>
<head><title>Search Employee</title></head>
```



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```
<body>
  <form action="EmployeeServlet" method="post">
    Enter Employee ID: <input type="text" name="empId">
    <input type="submit" value="Search">
  </form>
</body>
</html>
```

```
import java.io.*; import
javax.servlet.*;      import
javax.servlet.http.*; import
java.sql.*;
```

```
public class EmployeeServlet extends HttpServlet { protected void
doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {

    String empId = request.getParameter("empId");
    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    try {
        Class.forName("com.mysql.jdbc.Driver");
        Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/company", "root", "password");

        String query = "SELECT * FROM employees WHERE emp_id=?";
        PreparedStatement ps = con.prepareStatement(query);
        ps.setString(1, empId); ResultSet
        rs = ps.executeQuery();

        if (rs.next()) {
            out.println("<h2>Employee    Details</h2>");
            out.println("ID: " + rs.getInt(1) + "<br>");
            out.println("Name: " + rs.getString(2) + "<br>");
            out.println("Department: " + rs.getString(3));
        } else { out.println("No employee found with ID " +
            empId);
```

```
    }  
  
    con.close();  
} catch (Exception e) { out.println("Error:  
    " + e.getMessage());  
}  
}  
}
```

7.2.3 Output:

1) Enter an employee ID → Shows details if found.

2) Not found → "No employee found with ID X"

7.3.1 Aim: To develop a JSP-based student portal that accepts attendance data and saves it to the database using a servlet.

- Objective:**
- 1) Combine JSP for UI and Servlets for logic
 - 2) Perform INSERT using JDBC
 - 3) Build a real-world web flow

Code:

```
<%@ page language="java" %>  
<html>  
<head><title>Student Attendance</title></head>  
<body>  
    <h2>Mark Attendance</h2>  
    <form action="AttendanceServlet" method="post"> Roll  
        No: <input type="text" name="roll"><br>  
        Name: <input type="text" name="name"><br>  
        Status: <select name="status">  
            <option>Present</option>  
            <option>Absent</option>  
        </select><br>  
        <input type="submit" value="Submit">  
    </form>  
</body> </html>
```



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```
import java.io.*; import
javax.servlet.*; import
javax.servlet.http.*; import
java.sql.*;
```

```
public class AttendanceServlet extends HttpServlet { protected void
doPost(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException {
    String roll = request.getParameter("roll");
    String name = request.getParameter("name");
    String status = request.getParameter("status");

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    try {
        Class.forName("com.mysql.jdbc.Driver");
        Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/student_portal", "root",
"password");

        String query = "INSERT INTO attendance (roll_no, name, status) VALUES (?, ?,
?)";
        PreparedStatement ps = con.prepareStatement(query);
        ps.setString(1, roll); ps.setString(2, name);
        ps.setString(3, status);

        int i = ps.executeUpdate(); if
(i > 0) {
            out.println("<h3>Attendance marked successfully for " + name + "!</h3>");
        }

        con.close();
    } catch (Exception e) { out.println("Error:
" + e.getMessage());
    }
}
```

}

```
CREATE TABLE attendance ( id INT  
    AUTO_INCREMENT PRIMARY KEY,  
    roll_no VARCHAR(20),  
    name VARCHAR(100),  
    status VARCHAR(10)  
);
```

OUTPUT

Form submission → "Attendance marked successfully for John!" And the data is stored in the database.